## 4.5 Size of grinding machine

The size of a grinding machine is specified according to the size of the largest work piece that can be mounted on the machine.

Cylindrical grinding machine, is specified by the diameter and length of the largest workpiece that can be held.

Internal cylindrical grinding machine is specified by the maximum diameter of the workpiece that can be held in the chuck and maximum stroke length of the wheel.

Surface grinding machine, is specified by the table area and the maximum height of the grinding wheel from the table surface.

Rotary table or chuck type grinding machine is specified by the maximum diameter of the table or chuck. Tool and cutter grinder is specified by the maximum size of tool and cutters that can be sharpened and dressed.

## 4.6 Centreless grinding

Centreless grinding is a method of grinding external, cylindrical, tapered and formed surfaces on workpieces, the workpiece is not held between the centres or in chucks. It is placed in a floating condition between two grinding wheels. There are two types of centreless grinding and they are

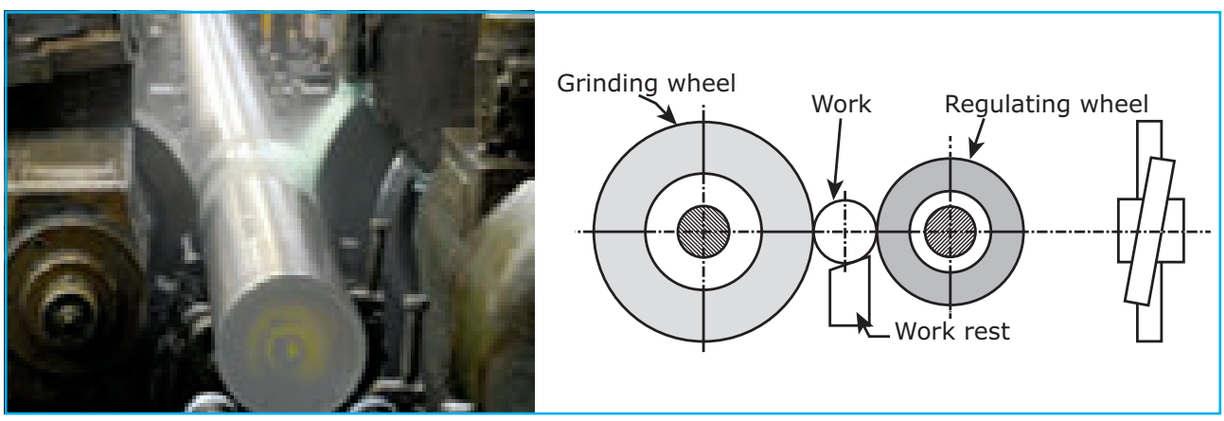
1. External centreless grinding
2. Internal centreless grinding

### External centreless grinding

Two wheels – a grinding and regulating wheel are used in external centreless grinding. Both wheels are rotated in the same direction. The work is placed upon the work rest and rotated between the wheels. The axial movement of the work pass through grinding wheel is obtained by tilting the regulating wheel at a slight angle from horizontal. An angular adjustment of 0 to 10 degrees provided in the machine for this purpose.

### Internal centreless grinding

The principles of external centreless grinding are applied to internal centreless grinding also. Grinding is done on the inner surface of the holes. In internal centreless grinding, the work is supported by three rolls – a regulating roll, a supporting roll and a pressure roll. The grinding wheel contacts the inside surface of the workpiece directly opposite the regulating roll. The distance between




External centreless Grinding



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